Decision Making in C (Selection Statements)

1. Introduction

In programming, **decision making** allows a program to choose between different actions depending on a condition.

In everyday life we make decisions such as:

- If it rains → carry an umbrella
- If marks ≥ 50 → the student passes
- If age ≥ 18 → eligible to vote

Similarly, a program can execute different instructions depending on whether a **condition is true or false**.

Decision making in C is implemented using **selection statements**.

Types of Decision Making Statements in C

C provides the following decision-making structures:

1. if statement
2. if-else statement
3. else-if ladder
4. Nested if statements
5. switch statement
6. Conditional (ternary) operator

1. The if Statement

The **if statement** executes a block of code **only when a specified condition is true**.

Syntax

```
if(condition)
{
   statements;
}
```

How it Works

1. The condition is evaluated.
2. If the condition is **true**, the statements inside the block execute.
3. If the condition is **false**, the block is skipped.

Example 1: Pass or Not

```
#include <stdio.h>
int main()
{
   int marks;
   printf("Enter marks: ");
   scanf("%d", &marks);
   if(marks >= 50)
   {
      printf("You passed the exam");
   }
   return 0;
}
```

Explanation

- If the marks are **50 or above**, the message is printed.
- If marks are **less than 50**, nothing happens.

2. The if-else Statement

The if-else statement allows a program to **choose between two alternatives**.

Syntax

```
if(condition)
{
   statements1;
}
else
{
   statements2;
}
```

How it Works

- If the condition is **true**, the first block executes.
- If the condition is **false**, the second block executes.

Example 2: Voting Eligibility

```
#include <stdio.h>

int main()
{
   int age;
   printf("Enter your age: ");
   scanf("%d", &age);
   if(age >= 18)
   {
      printf("You are eligible to vote");
   }
   else
   {
      printf("You are not eligible to vote");
   }

   return 0;

}
```

3. The else-if Ladder

When there are **multiple conditions**, the else-if ladder is used.

Syntax

```
if(condition1)
{
   statements;
}
else if(condition2)
{
   statements;
}
else if(condition3)
{
   statements;
}
else
{
   statements;
}
```

The program checks conditions **from top to bottom** and executes the **first true condition**.

Example 3: Student Grading System

```
#include <stdio.h>
int main()
{
   int marks;
   printf("Enter marks: ");
   scanf("%d", &marks);
   if(marks >= 80)

      printf("Grade A");

   else if(marks >= 70)

      printf("Grade B");
   else if(marks >= 60)
      printf("Grade C");
   else if(marks >= 50)
```

```
        printf("Grade D");
    else
        printf("Fail");

    return 0;
}
```

4. Nested if Statements

A **nested if** is an if statement placed inside another if.

It is used when **multiple conditions must be checked sequentially**.

Example: University Admission

```
#include <stdio.h>

int main()
{
    int marks;
    int interview;

    printf("Enter marks: ");
    scanf("%d", &marks);

    if(marks >= 70)
    {
        printf("Eligible for interview\n");

        printf("Enter interview score: ");
        scanf("%d", &interview);

        if(interview >= 50)
        {
            printf("Admission granted");
        }
    }
    return 0;
}
```

Explanation:

1. Student must score **70 marks or more**.
2. Then the **interview score is checked**.

5. The switch Statement

The **switch statement** is used when a variable can have **many possible values**, and we want to execute different blocks depending on those values.

Syntax

```
switch(expression)
{
   case value1:
      statements;
      break;

   case value2:
      statements;
      break;

   case value3:
      statements;
      break;

   default:
      statements;
}
```

Important Components

Keyword	**Meaning**
switch	Tests the value
case	Represents a possible value
break	Stops execution
default	Executes if no case matches

Example 4: Day of the Week

```
#include <stdio.h>
int main()
{
   int day;
   printf("Enter a number (1-7): ");
   scanf("%d", &day);
   switch(day)
   {
      case 1:
         printf("Monday");
         break;
      case 2:
         printf("Tuesday");
         break;
      case 3:
         printf("Wednesday");
         break;

      case 4:
         printf("Thursday");
         break;
      case 5:
         printf("Friday");
         break;
      case 6:
         printf("Saturday");
         break;
      case 7:
         printf("Sunday");
         break;
      default:
         printf("Invalid day");
   }
   return 0;
}
```

Importance of break

Without break, the program will continue executing the next cases. This is called **fall-through**.

Example:

```
case 1:
printf("Monday");
case 2:
printf("Tuesday");
```

If day = 1, the program prints:

Monday

Tuesday

6. Conditional (Ternary) Operator

The **ternary operator** is a shorter form of if-else.

Syntax

condition ? expression1 : expression2;

Meaning:

- If condition is **true → expression1**
- If condition is **false → expression2**

Example

```
int a = 10;

int b = 20;

int max = (a > b) ? a : b;

printf("Maximum = %d", max);
```

Output

Maximum = 20

Comparison: if vs switch

Feature	if	switch
Condition	Any condition	Exact values
Operators	Relational/logical	Equality only
Use case	Complex conditions	Many fixed choices

Common Mistakes Made

1. Using = instead of ==

Wrong:

```
if(a = 10)
```

Correct:

```
if(a == 10)
```

2. Missing break in switch

This causes unintended outputs.

3. Incorrect condition order

Example:

Wrong:

```
if(marks >= 50)
printf("Pass");
else if(marks >= 80)
printf("Grade A");
```

Correct:

```
if(marks >= 80)
printf("Grade A");
else if(marks >= 50)
printf("Pass");
```